Scale-invariant propagated class exclusion for direct semi-supervised clustering

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Abstract

Semi-supervised non-negative matrix factorization (NMF) usually transfers labels by fitting latent coordinates or smoothing neighboring representations. These strategies specify what an unlabeled representation should resemble. We study the complementary problem of determining which class contributions should be suppressed. Propagated Class-Exclusion NMF (PCE-NMF) converts harmonic class evidence into sample-specific class-incompatibility weights and penalizes the corresponding rank-one contribution energies. The penalty and the induced direct clustering rule are invariant under the diagonal scaling ambiguity of NMF. We establish the harmonic-evidence properties, derive an equivalent canonical shrinkage model, and prove monotonic descent of the objective under the stated alternating updates. Experiments use six benchmark data sets, 20 paired random seeds, identical labeled splits, and evaluation restricted to unlabeled samples. Among eleven compared methods, PCE-NMF ranks within the top three for every data-set–metric combination and obtains the highest mean accuracy and normalized mutual information on COIL20 and COIL100. It achieves higher mean accuracy and normalized mutual information than HSSNMF, CSNMF, and SPNMF on all six data sets, with all 36 paired comparisons significant after within-baseline Holm correction. The controlled ablation further shows substantial gains over hard class exclusion in every paired trial. The results establish propagated class exclusion as an effective direct-clustering mechanism, especially under limited supervision in multi-class object data.

Keywords: non-negative matrix factorization; semi-supervised clustering; label propagation; class exclusion; direct clustering; scale invariance

1 Introduction

Non-negative matrix factorization (NMF) has become a widely used tool for low-dimensional representation and clustering because its non-negativity constraints lead to an additive and often interpretable representation of non-negative data [1, 2]. Given a non-negative data matrix $X \in \mathbb{R}_+^{m\times n}$, standard NMF seeks two non-negative factor matrices whose product approximates X. The learned coefficient matrix can reveal latent clustering structure, but standard NMF does not use available class labels.

Semi-supervised NMF incorporates partial labels into the factorization. Existing methods mainly follow several representative strategies. Constraint-based methods enforce label-consistent structures in the latent representation [3]. Discriminative methods impose penalties that encourage labeled samples to exhibit class-consistent representations [4, 5]. More recently, direct constraint models avoid fitting a prescribed label vector. For example, GOCNMF fixes labeled representations to one-hot vectors and transfers this structure through graph regularization [6], whereas LCNMF sets only class-inconsistent coordinates of labeled representations to zero and leaves their class-consistent coordinates adaptive [7].

Another line of research propagates labels or pairwise constraints through sample graphs. Constraint-propagation NMF modifies pairwise relations [8]; HSSNMF propagates constraints through a hypergraph [9]; SPNMF jointly propagates pointwise and pairwise supervision [10]; and weighted or high-order label propagation regularizes latent representations or label-prediction variables [11, 12, 13]. Self-supervised and robust semi-supervised NMF models further enrich the supervisory signal or the reconstruction loss [14]. Across these formulations, supervision specifies target coordinates, sample relations, or neighborhood agreement.

This work considers a different question. Instead of asking whether an unlabeled representation should be close to a propagated label vector, we ask whether propagated class evidence can be used to suppress *class-incompatible NMF contributions*. Let P_{ik} denote the propagated evidence that sample i belongs to class k. A large value of P_{ik} indicates that the kth class contribution should be retained, whereas a small value indicates incompatibility and should induce stronger suppression. To implement this principle without introducing dependence on the arbitrary column scaling of NMF, we use the scale-invariant contribution energy

$$\|u_k\|_2^2 v_{ik}^2,$$

where u_k is the kth basis vector and v_{ik} is the corresponding coefficient of sample i.

Based on this observation, we propose *Propagated Class-Exclusion Non-negative Matrix Factorization* (PCE-NMF). Harmonic propagation first produces a class-evidence matrix from the available labels. Its complement defines a sample-specific class-incompatibility field. PCE-NMF then penalizes the contribution energy of each class according to this incompatibility, while labeled samples retain an explicit hard exclusion of their class-inconsistent coordinates. In this way, supervision is expressed as class-dependent shrinkage rather than absolute one-hot fitting or pairwise graph smoothing.

The propagated penalty respects the standard diagonal scaling ambiguity of NMF. Indeed, under the transformation $U \mapsto US$ and $V \mapsto VS^{-1}$ with a positive diagonal matrix S, both the reconstruction and every supervised contribution

remain unchanged. This property leads to a canonical interpretation of PCE-NMF: after normalizing each basis column, the proposed regularizer becomes a sample- and class-dependent quadratic shrinkage on the canonical coefficient matrix. The supervised quantity is therefore independent of arbitrary positive diagonal rescaling of the factors.

The main contributions of this work are summarized as follows.

1. We introduce propagated class exclusion, which converts harmonic class evidence into sample-specific suppression of class-incompatible rank-one contributions.
2. We prove the harmonic-evidence bounds and diagonal-scale invariance, and derive an equivalent canonical shrinkage representation.
3. We derive multiplicative updates with monotonic objective descent and a direct class-energy decision rule built from the same invariant quantity.
4. Paired experiments separate the contribution of propagated exclusion from hard support constraints and characterize its behavior across label rates, graph neighborhoods, data sets, and competing NMF mechanisms.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the proposed PCE-NMF model and its direct clustering rule. Section 4 presents the theoretical analysis, optimization algorithm, and computational complexity. Section 5 reports the experimental results, and Section 6 concludes the paper.

2 Related Work

2.1 Label-constrained and discriminative NMF

A direct way of incorporating partial labels into NMF is to constrain the latent representation. Constrained NMF (CNMF) constructs a label-constraint matrix so that labeled samples from the same class share related low-dimensional representations [3]. Subsequent methods use soft discriminative penalties instead of an exact structural constraint. For example, discriminative semi-supervised NMF methods encourage the representations of labeled samples to be consistent with their known labels while retaining reconstruction flexibility [4].

More recent direct-clustering formulations further weaken the requirement that a labeled representation must match a prescribed one-hot vector. EDNMF constructs discriminative regularizers from the relative magnitudes of class-consistent and class-inconsistent elements of labeled representations [5]. LCNMF instead fixes the class-inconsistent coordinates of each labeled representation to zero while leaving its class-consistent coordinate as a learnable variable [7]. These formulations improve representation flexibility and allow cluster assignments to be read directly from the learned coefficients.

GOCNMF adopts a different direct strategy by fixing labeled representations to their one-hot label vectors and using graph regularization to guide the unlabeled representations [6]. If the labeled samples are placed first, its representation matrix has the form

$$\widetilde{V} = \begin{bmatrix} C \\ V_{\mathcal{U}} \end{bmatrix},$$

where C is the one-hot label matrix. The graph term then encourages neighboring samples to have similar latent representations. Such a design provides strong class semantics, but the supervision is imposed directly on the latent coordinates.

The proposed PCE-NMF differs from these approaches in the object on which supervision is imposed. For labeled samples, PCE-NMF retains a hard class-exclusion structure similar in spirit to local coordinate constraints. For unlabeled samples, propagated supervision acts on the scale-invariant class contribution $\|u_k\|_2^2 v_{ik}^2$ through a sample-specific class-incompatibility weight.

2.2 Propagation-based semi-supervised NMF

Label propagation provides an effective mechanism for transferring limited supervision from labeled to unlabeled samples. Several semi-supervised NMF methods have therefore coupled matrix factorization with label propagation, pairwise-constraint propagation, or adaptive graph learning [8, 11, 10].

Recent Pattern Recognition models illustrate complementary propagation routes. HSSNMF propagates pairwise constraints through a hypergraph and uses the resulting similarity structure in symmetric NMF [9]. SPNMF propagates both labels and pairwise constraints and combines them with pivotal-aware weighting [10]. CSNMF addresses a different axis by combining pointwise and pairwise supervision with a correntropy loss for contaminated data [14]. These models expand the amount, order, or robustness of the supervisory information used by NMF.

SNMFWLP integrates weighted label propagation directly into NMF and uses the low-dimensional representation itself as a label-prediction variable [12]. Its representative objective contains both a graph smoothness term and a weighted fitting term,

$$\begin{aligned} &\|X - WH^T\|_F^2 + \alpha \operatorname{Tr}(H^T L H) \\ &\quad + \beta \operatorname{Tr}\big((H-M)^T \mathcal{U} (H-M)\big), \end{aligned}$$

together with an orthogonality constraint on the basis matrix. Thus, propagated supervision explicitly pulls the learned representation toward a weighted label target.

Other recent models enrich the propagation mechanism itself. Positive and negative label propagation can be combined with constraint propagation to refine neighborhood relations [11], while high-order propagation methods exploit multi-hop relations beyond first-order graph neighbors [13]. These developments demonstrate that propagation can substantially enhance semi-supervised NMF, but their supervision is still primarily expressed through representation fitting, pairwise relations, or propagation-induced graph regularization.

PCE-NMF uses propagation differently. The harmonic solution is computed before factorization and is converted into a fixed class-incompatibility field. The resulting penalty is

diagonal with respect to sample-class coordinates:

$$\sum_{i,k}(1-P_{ik})\|u_k\|_2^2 v_{ik}^2.$$

Hence, unlike a Laplacian term, it introduces no cross-sample product $v_i^T v_j$. Unlike label fitting, it does not force v_i to approximate a prescribed vector. Its role is instead to attenuate class contributions that are inconsistent with the propagated evidence.

2.3 Position of the proposed method

The distinction among representative semi-supervised NMF strategies can be summarized by the location and form of supervision. Label-fitting methods specify desirable latent coordinates, discriminative methods compare class-consistent and class-inconsistent coordinates, and graph-based methods couple representations of neighboring samples. PCE-NMF occupies a distinct mechanism-level position: propagated evidence determines the strength of a class-wise exclusion penalty on scale-invariant contribution energies.

This distinction is important because standard NMF admits the diagonal transformation

$$U \mapsto US, \qquad V \mapsto VS^{-1},$$

for any positive diagonal S. A supervisory quantity based solely on a raw coefficient may therefore depend on an arbitrary factor scaling unless an additional normalization or structural constraint is imposed. In PCE-NMF, the product $\|u_k\|_2^2 v_{ik}^2$ is invariant under this transformation. The proposed supervision consequently has a direct interpretation that is independent of the NMF column gauge.

3 Propagated Class-Exclusion NMF

Figure 1 summarizes the two coupled routes of PCE-NMF. The propagation route converts partial labels and the cosine p-nearest-neighbor graph into harmonic class evidence P and class-exclusion weights Q. The factorization route learns U and V under hard labeled-support constraints. The two routes are coupled through the class-specific contribution energy $s_{ik} = \|u_k\|_2^2 v_{ik}^2$, which is regularized by q_{ik} and used directly for cluster assignment.

3.1 Notation and standard NMF

Let

$$X = [x_1, \ldots, x_n] \in \mathbb{R}_+^{m\times n}$$

contain n non-negative samples from c classes. Let $\mathcal{L}$ and $\mathcal{U}$ denote the labeled and unlabeled index sets, respectively, with $\mathcal{L}\cap\mathcal{U} = \varnothing$ and $\mathcal{L}\cup\mathcal{U} = \{1,\ldots,n\}$.

We consider a rank-c factorization

$$X \approx UV^T,$$

where

$$U = [u_1, \ldots, u_c] \in \mathbb{R}_+^{m\times c}, \qquad V = [v_{ik}] \in \mathbb{R}_+^{n\times c}.$$

Standard Euclidean NMF minimizes

$$\min_{U,V\geq 0} \|X - UV^T\|_F^2. \tag{1}$$

For a labeled sample $i \in \mathcal{L}$, let $y_i \in \{1,\ldots,c\}$ denote its class label. Let

$$C \in \{0,1\}^{|\mathcal{L}|\times c}$$

be the corresponding one-hot label matrix.

3.2 Harmonic class evidence

All samples are normalized to unit Euclidean norm before graph construction. For each sample x_i, let $\mathcal{N}_p(i)$ contain the p other samples x_j, $j \neq i$, with the largest cosine similarities $x_i^T x_j$. The self-similarity is excluded before neighbor selection. Ties are resolved by the ascending sample index returned by MATLAB's stable ordering. The binary union graph is

$$W_{ij} = \begin{cases} 1, & j \in \mathcal{N}_p(i) \text{ or } i \in \mathcal{N}_p(j), \\ 0, & \text{otherwise}, \end{cases} \qquad W_{ii} = 0. \tag{2}$$

Thus $W \in \mathbb{R}_+^{n\times n}$ is symmetric and contains no self-loop. Let

$$D = \operatorname{diag}(W\mathbf{1}), \qquad L = D - W$$

be its degree matrix and graph Laplacian. Class evidence is obtained from the Gaussian-field harmonic solution [15].

Because disconnected components without labeled samples cannot receive reliable class evidence, harmonic propagation is performed separately on each connected component. Consider a component containing at least one labeled node, and partition its vertices into labeled and unlabeled sets. The propagated class-evidence matrix is defined by

$$P_{\mathcal{L}} = C, \qquad L_{\mathcal{U}\mathcal{U}} P_{\mathcal{U}} = W_{\mathcal{U}\mathcal{L}} C. \tag{3}$$

For every unlabeled sample contained in an anchored component, P_{ik} represents the propagated evidence of class k. Let $\mathcal{A} \subseteq \mathcal{U}$ denote the set of such anchored unlabeled samples. We define the class-incompatibility field

$$q_{ik} = \begin{cases} 1 - P_{ik}, & i \in \mathcal{A}, \\ 0, & i \in \mathcal{L} \text{ or } i \in \mathcal{U}\setminus\mathcal{A}. \end{cases} \tag{4}$$

Thus, high class evidence yields weak exclusion, whereas low class evidence yields strong exclusion. Setting $q_{ik} = 0$ on unanchored components avoids introducing artificial supervision when no labeled information can reach the sample.

3.3 Hard exclusion for labeled samples

For each labeled sample, we impose

$$v_{ik} = 0, \qquad i \in \mathcal{L}, \quad k \neq y_i. \tag{5}$$

The class-consistent coefficient $v_{i\,y_i}$ remains learnable. Therefore, the known label determines only the support pattern of the labeled representation and does not impose a fixed amplitude.

3.4 The PCE-NMF model

For sample i and class k, define the class-specific contribution energy

$$s_{ik} = \|u_k\|_2^2 v_{ik}^2. \tag{6}$$

Table 1: Mechanism-level position of representative semi-supervised NMF methods. "Direct" indicates that the learned class-semantic coordinates are used for assignment without an additional classifier.

Method	Propagated object	Supervision acts on	Primary operation	Direct	Explicit invariant score
CNMF	none	latent coordinates	equality structure	no	not reported
CSNMF	pairwise constraints	loss and representation	robust fitting	no	not reported
HSSNMF	pairwise constraints	similarity and assignment	hypergraph coupling	no	not reported
SPNMF	labels and constraints	representation and relations	dual propagation	no	not reported
GOCNMF	graph neighborhoods	latent coordinates	one-hot smoothing	yes	not reported
SNMFWLP	weighted labels	latent coordinates	target fitting	no	not reported
PCE-NMF	class evidence	rank-one contributions	incompatibility shrinkage	yes	yes

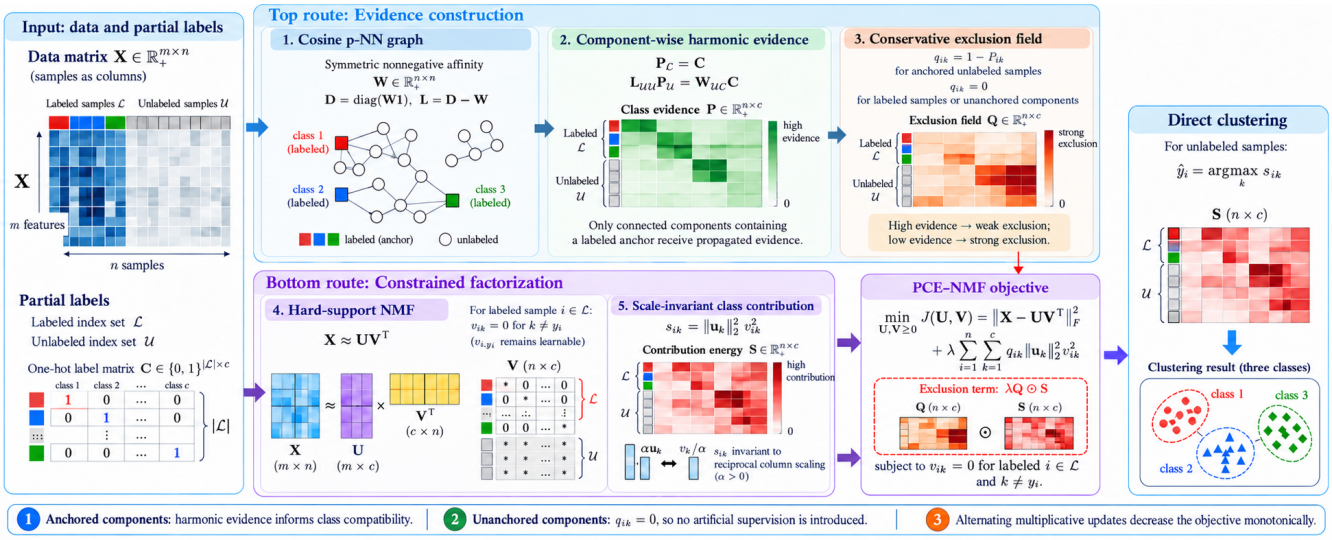


Figure 1: PCE-NMF from harmonic class evidence to conservative class exclusion. Partial labels are propagated over the sample graph to obtain harmonic class evidence, which is converted into conservative class-exclusion weights. These weights regularize the scale-invariant contribution energies of the NMF model, enabling direct cluster assignment.

This quantity is the squared Frobenius norm of the rank-one term $u_k v_{ik}$ in the reconstruction of sample i. The complete reconstruction energy also contains cross-products between different rank-one terms; the score in (6) measures the magnitude of one class contribution. The proposed PCE-NMF model is

$$\min_{U,V \geq 0} J(U,V) = \|X - UV^T\|_F^2 + \lambda \sum_{i=1}^{n} \sum_{k=1}^{c} q_{ik} \|u_k\|_2^2 v_{ik}^2, \tag{7}$$

subject to (5), where $\lambda > 0$ controls the overall strength of propagated exclusion.

The second term in (7) has a direct interpretation. If P_{ik} is large, then $q_{ik} = 1 - P_{ik}$ is small and the kth contribution of sample i is weakly penalized. If P_{ik} is small, the same contribution receives stronger shrinkage. The model therefore does not force the coefficient vector v_i to reproduce P_i. Instead, propagated evidence modulates how strongly each class-specific NMF contribution is allowed to remain active.

When $\lambda = 0$, (7) reduces to Hard Class-Exclusion NMF (HCE-NMF), which retains the hard support constraint (5) and removes propagated exclusion.

3.5 Direct clustering

Because the score in (6) measures a class-specific contribution in a scale-invariant form, PCE-NMF produces cluster assignments directly:

$$\widehat{y}_i = \arg \max_{1 \leq k \leq c} \|u_k\|_2^2 v_{ik}^2. \tag{8}$$

4 Theoretical Analysis

We next establish the main structural properties of PCE-NMF and derive its multiplicative optimization scheme.

4.1 Validity of harmonic class evidence

Lemma 1 (Validity of harmonic class evidence). Consider a connected graph component containing at least one labeled vertex. The harmonic solution in (3) satisfies

$$P_\mathcal{U} \geq 0, \qquad P_\mathcal{U} \mathbf{1}_c = \mathbf{1}_{|\mathcal{U}|}.$$

Consequently,

$$0 \leq P_{ik} \leq 1$$

for every unlabeled vertex in the component, and hence $0 \leq q_{ik} \leq 1$.

Proof. For a connected component containing at least one labeled vertex, the Dirichlet Laplacian $L_{\mathcal{U}\mathcal{U}}$ is a nonsingular M-matrix. Therefore

$$L_{\mathcal{U}\mathcal{U}}^{-1} \geq 0.$$

Since $W_{\mathcal{U}\mathcal{L}}C \geq 0$, equation (3) gives

$$P_{\mathcal{U}} = L_{\mathcal{U}\mathcal{U}}^{-1} W_{\mathcal{U}\mathcal{L}} C \geq 0.$$

Moreover, because the rows of the full graph Laplacian sum to zero,

$$L_{\mathcal{U}\mathcal{U}} \mathbf{1}_{|\mathcal{U}|} = W_{\mathcal{U}\mathcal{L}} \mathbf{1}_{|\mathcal{L}|}.$$

Since $C\mathbf{1}_c = \mathbf{1}_{|\mathcal{L}|}$,

$$P_{\mathcal{U}} \mathbf{1}_c = L_{\mathcal{U}\mathcal{U}}^{-1} W_{\mathcal{U}\mathcal{L}} C \mathbf{1}_c = L_{\mathcal{U}\mathcal{U}}^{-1} W_{\mathcal{U}\mathcal{L}} \mathbf{1}_{|\mathcal{L}|} = \mathbf{1}_{|\mathcal{U}|}.$$

Non-negativity and unit row sums imply $0 \leq P_{ik} \leq 1$. The result for q_{ik} follows from (4). □

4.2 Scale invariance and canonical representation

Theorem 1 (Scale invariance)**.** *Let*

$$S = \mathrm{diag}(s_1, \ldots, s_c), \qquad s_k > 0,$$

and define

$$\widetilde{U} = US, \qquad \widetilde{V} = VS^{-1}.$$

Then

$$J(\widetilde{U}, \widetilde{V}) = J(U, V).$$

Furthermore, the hard-support constraints in (5) *are preserved.*

Proof. The reconstruction is unchanged because

$$\widetilde{U}\widetilde{V}^T = US(VS^{-1})^T = UV^T.$$

For each class k,

$$\|\widetilde{u}_k\|_2^2 \widetilde{v}_{ik}^2 = s_k^2 \|u_k\|_2^2 \frac{v_{ik}^2}{s_k^2} = \|u_k\|_2^2 v_{ik}^2.$$

Hence every term of the propagated-exclusion penalty is unchanged. Finally, $v_{ik} = 0$ if and only if $\widetilde{v}_{ik} = 0$, so the feasible support pattern is preserved. □

Corollary 1 (Scale-invariant direct decision)**.** *Under the transformation in Theorem 1,*

$$\arg\max_k \|\widetilde{u}_k\|_2^2 \widetilde{v}_{ik}^2 = \arg\max_k \|u_k\|_2^2 v_{ik}^2.$$

Therefore, the direct clustering rule (8) *is independent of the standard diagonal scaling ambiguity of NMF.*

Proposition 1 (Canonical shrinkage representation)**.** *Assume* $\|u_k\|_2 > 0$ *for* $k = 1, \ldots, c$*. Define*

$$d_k = \|u_k\|_2, \qquad D_u = \mathrm{diag}(d_1, \ldots, d_c),$$

and

$$B = UD_u^{-1}, \qquad Z = VD_u.$$

Then $\|b_k\|_2 = 1$*,* $UV^T = BZ^T$*, and PCE-NMF is equivalently written as*

$$\min_{B,Z \geq 0} \|X - BZ^T\|_F^2 + \lambda \sum_{i=1}^{n} \sum_{k=1}^{c} q_{ik} z_{ik}^2, \tag{9}$$

subject to $\|b_k\|_2 = 1$ *and the corresponding hard-support constraints.*

Proof. By construction,

$$\|b_k\|_2 = \frac{\|u_k\|_2}{d_k} = 1,$$

and

$$BZ^T = UD_u^{-1}(VD_u)^T = UV^T.$$

Moreover,

$$z_{ik} = d_k v_{ik},$$

so

$$z_{ik}^2 = \|u_k\|_2^2 v_{ik}^2.$$

Substituting these identities into (7) gives (9). □

Proposition 1 shows that the propagated-exclusion term can be interpreted as a sample- and class-dependent quadratic shrinkage on a canonical coefficient representation. The class semantics of this penalty are therefore not created by an arbitrary column scaling of the factorization. The canonical form is defined on nondegenerate factor pairs whose basis columns are all nonzero.

4.3 Evidence-exclusion relation

For an anchored unlabeled sample i, define

$$s_i = (s_{i1}, \ldots, s_{ic})^T, \qquad S_i = \sum_{k=1}^{c} s_{ik},$$

and let

$$R_i = \sum_{k=1}^{c} (1 - P_{ik}) s_{ik}$$

denote its propagated-exclusion penalty before multiplication by λ.

Proposition 2 (Evidence-exclusion decomposition)**.** *For every anchored unlabeled sample,*

$$R_i = S_i - P_i^T s_i. \tag{10}$$

Moreover, if

$$P_i^{\max} = \max_k P_{ik}, \qquad P_i^{\min} = \min_k P_{ik},$$

then

$$(1 - P_i^{\max}) S_i \leq R_i \leq (1 - P_i^{\min}) S_i. \tag{11}$$

Proof. Using $\sum_k P_{ik} = 1$ from Lemma 1,

$$\begin{aligned} R_i &= \sum_k (1 - P_{ik}) s_{ik} = \sum_k s_{ik} - \sum_k P_{ik} s_{ik} \\ &= S_i - P_i^T s_i. \end{aligned}$$

Because $s_{ik} \geq 0$,

$$P_i^{\min} S_i \leq P_i^T s_i \leq P_i^{\max} S_i.$$

Subtracting these inequalities from S_i gives (11). □

For fixed total contribution energy S_i, minimizing R_i is therefore equivalent to maximizing $P_i^T s_i$. This provides an interpretation of PCE-NMF as aligning class-specific contribution energy with propagated class evidence without explicitly fitting s_i to P_i.

Corollary 2 (Uniform-evidence neutrality)**.** *If*

$$P_{ik} = \frac{1}{c}, \qquad k = 1, \dots, c,$$

then

$$R_i = \left(1 - \frac{1}{c}\right) S_i.$$

Thus, uniform propagated evidence does not favor any class and induces only a class-independent shrinkage of the total contribution energy.

4.4 Multiplicative updating rules

Let

$$a_k = \|u_k\|_2^2, \qquad b_k = \sum_{i=1}^{n} q_{ik} v_{ik}^2.$$

The gradient of (7) with respect to U is

$$\nabla_U J = 2U(V^T V) - 2XV + 2\lambda U \operatorname{diag}(b_1, \dots, b_c).$$

Separating positive and negative parts gives the multiplicative rule

$$U \leftarrow U \odot \frac{XV}{U(V^T V) + \lambda U \operatorname{diag}(b_1, \dots, b_c)}. \tag{12}$$

Similarly,

$$\nabla_V J = 2V(U^T U) - 2X^T U + 2\lambda (Q \odot V) \operatorname{diag}(a_1, \dots, a_c),$$

where $Q = [q_{ik}]$. Hence every free coefficient is updated by

$$V \leftarrow V \odot \frac{X^T U}{V(U^T U) + \lambda (Q \odot V) \operatorname{diag}(a_1, \dots, a_c)}. \tag{13}$$

After each V update, the constrained entries in (5) remain fixed at zero.

Numerical implementation. The MATLAB R2019a implementation initializes every free entry by a positive common factor plus 10^{-2} and evaluates each multiplicative denominator as $\max\{d, 10^{-12}\}$ elementwise. After an update, free entries are clipped below at 10^{-12}, and labeled wrong-class entries are reset exactly to zero. The graph and harmonic evidence remain fixed during factorization. All reported experiments use 50 PCE-NMF iterations. These operations specify the executed finite-precision algorithm; the theorem below analyzes the corresponding ideal multiplicative updates.

Proposition 3 (Evidence-monotone one-step shrinkage)**.** *Fix the current U and V, and consider an anchored unlabeled entry $v_{ik} > 0$ for which the numerator and denominator in* (13) *are strictly positive. In one application of* (13)*, the updated value v_{ik}^+ is a non-decreasing function of P_{ik}.*

Proof. For fixed current factors, (13) can be written as

$$v_{ik}^+ = v_{ik} \frac{(X^T U)_{ik}}{[V(U^T U)]_{ik} + \lambda(1 - P_{ik}) \|u_k\|_2^2 v_{ik}}.$$

Let

$$A = [V(U^T U)]_{ik}, \quad B = \|u_k\|_2^2 v_{ik}, \quad C = v_{ik}(X^T U)_{ik}.$$

Since $A, B, C \geq 0$,

$$\frac{\partial v_{ik}^+}{\partial P_{ik}} = \frac{\lambda BC}{[A + \lambda(1 - P_{ik})B]^2} \geq 0.$$

Thus, greater propagated evidence results in weaker one-step shrinkage of the corresponding class coefficient. □

Proposition 3 characterizes the immediate response of one coefficient update to its propagated class evidence at fixed factors.

4.5 Monotonicity of the objective

We use the following standard majorization fact.

Lemma 2. Let A be symmetric and entrywise non-negative, and let $\bar{z} > 0$. Then

$$K(\bar{z}) = \operatorname{diag}\left(\frac{(A\bar{z})_1}{\bar{z}_1}, \dots, \frac{(A\bar{z})_r}{\bar{z}_r}\right)$$

satisfies

$$K(\bar{z}) - A \succeq 0.$$

Proof. For any $x \in \mathbb{R}^r$,

$$x^T (K(\bar{z}) - A)\, x = \frac{1}{2} \sum_{p,q} A_{pq} \left(x_p \sqrt{\frac{\bar{z}_q}{\bar{z}_p}} - x_q \sqrt{\frac{\bar{z}_p}{\bar{z}_q}}\right)^2 \geq 0.$$

This completes the proof. □

Theorem 2 (Monotonic decrease of PCE-NMF)**.** *Assume that, at every iteration, all free entries and all numerators and denominators in the ideal multiplicative updates* (12)*–*(13) *are strictly positive. Apply these updates in exact arithmetic while keeping the hard-support zeros fixed. Then*

$$J(U^{(t+1)}, V^{(t+1)}) \leq J(U^{(t)}, V^{(t)}).$$

Consequently, the sequence of objective values converges.

Proof. First fix V. Let

$$b_k = \sum_i q_{ik} v_{ik}^2$$

and define

$$A_U = V^T V + \lambda \operatorname{diag}(b_1, \dots, b_c).$$

Up to terms independent of U,

$$J(U, V) = \operatorname{Tr}(U A_U U^T) - 2 \operatorname{Tr}(XVU^T) + \text{const}.$$

The matrix A_U is symmetric and entrywise non-negative. Applying Lemma 2 row by row yields the standard diagonal auxiliary-function majorizer for this non-negative quadratic problem. Minimizing the majorizer gives (12). Hence the U update does not increase J.

Next fix U. For sample i, define

$$A_i = U^T U + \lambda \operatorname{diag}\big(q_{i1} a_1, \dots, q_{ic} a_c\big), \quad a_k = \|u_k\|_2^2.$$

Up to constants independent of v_i,

$$J_i(v_i) = v_i^T A_i v_i - 2 x_i^T U v_i.$$

Again, A_i is symmetric and entrywise non-negative. Restricting to the free coordinates of v_i, its corresponding principal submatrix retains these properties. Lemma 2 therefore gives a separable quadratic auxiliary function whose minimizer is exactly the free-coordinate update in (13). The fixed hard-support zeros are not altered. Thus the V update also cannot increase J.

Alternating the two updates yields

$$J(U^{(t+1)}, V^{(t+1)}) \leq J(U^{(t)}, V^{(t)}).$$

Finally, every term in (7) is non-negative, so $J \geq 0$. Therefore the non-increasing sequence of objective values converges. □

Remark 1. Theorem 2 establishes monotonic descent and convergence of the scalar objective sequence on the positive free-coordinate domain. Factor convergence and stationarity require additional assumptions. The empirical convergence study reports the finite-precision objective trajectories generated by the implemented updates.

4.6 Computational complexity

After the harmonic evidence has been computed, one PCE-NMF iteration mainly requires the products XV, X^TU, V^TV, and U^TU. The dominant factorization cost is therefore

$$O\big(mnc + (m+n)c^2\big)$$

per iteration. Computing the propagated-exclusion weights and their diagonal loading requires only $O(nc)$ additional operations, apart from the class-evidence propagation stage.

The harmonic propagation stage solves sparse Dirichlet systems with c right-hand sides on the anchored graph components. If $\mathcal{S}(L_{\mathcal{U}\mathcal{U}}, c)$ denotes the cost of the corresponding sparse solve, this cost is incurred once before the NMF iterations and no graph variable is updated during factorization. Constructing an exact cosine-neighbor graph from dense data requires $O(mn^2)$ arithmetic and $O(n^2)$ temporary storage in a direct implementation. The released code uses blocked neighbor search and stores the final graph sparsely, reducing the working memory while leaving the worst-case arithmetic order unchanged. The cost and fill-in of the component-wise Dirichlet solves depend on graph sparsity and ordering; these one-time graph operations can dominate the NMF iterations when n is large.

5 Experiments

The experiments address four questions. First, does PCE-NMF improve direct semi-supervised clustering under a paired protocol? Second, what performance is contributed by propagation beyond hard labeled support constraints? Third, how do label rate and graph neighborhood size change the result? Fourth, do the implemented updates preserve the predicted descent and support structure?

5.1 Data sets and preprocessing

Six benchmark data sets are used: YaleB, COIL20, Optdigits, ORL, PIE, and COIL100. Their dimensions and graph settings are listed in Table 2. The data include face images, object images, and handwritten digits, and vary from 400 to 7200 samples and from 10 to 100 classes.

Table 2: Data sets and graph neighborhood sizes. Here, p_{PCE}, p_{GOC}, and p_{SN} denote the neighborhood sizes used by PCE-NMF, GOCNMF, and SNMFWLP, respectively. GNMFLD uses $p = 5$ throughout.

Data set	Features	Samples	Classes	p_{PCE}	p_{GOC}	p_{SN}
YaleB	1024	2414	38	2	2	3
COIL20	1024	1440	20	3	3	3
Optdigits	64	5620	10	4	4	4
ORL	4096	400	40	3	3	3
PIE	1024	2856	68	3	3	3
COIL100	1024	7200	100	3	3	3

For every method, samples are represented as columns and normalized to unit Euclidean norm. No centering, dimensionality reduction, balancing, or data-set subsampling is applied. For each class k, the labeled subset contains

$$\max\{1, \lfloor 0.1 n_k \rfloor\} \tag{14}$$

randomly selected samples, where n_k is the number of samples in that class. All remaining samples are treated as unlabeled. We use the 20 paired random seeds

$$20260617, \ldots, 20260636. \tag{15}$$

Within each data-set–seed pair, all methods receive the same labeled split. Paired non-negative initial factors are also shared whenever permitted by the factorization structure.

Accuracy (ACC) and normalized mutual information (NMI) are computed only on the unlabeled index set $\mathcal{U}$. Let $\widehat{y}_i$ denote a predicted cluster index and let Π_c be the permutations of c labels. The external comparison uses

$$\mathrm{ACC} = \max_{\pi \in \Pi_c} \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \mathbf{1}\{y_i = \pi(\widehat{y}_i)\}, \tag{16}$$

$$\mathrm{NMI} = \frac{I(Y; \widehat{Y})}{\sqrt{H(Y) H(\widehat{Y})}}. \tag{17}$$

The optimal permutation is an evaluation-only relabeling of the predicted partition and is computed from the unlabeled evaluation set. NMF, CNMF, HSSNMF, CSNMF, and SPNMF use 20-start K-means. Each start runs at most 100 Lloyd iterations; an empty cluster is reseeded with the sample having the largest current assignment distance, and the solution with the smallest within-cluster sum of squares is retained. GNMFLD, EDNMF, LCNMF, GOCNMF, SNMFWLP, and PCE-NMF use their native row-wise assignments. PCE-NMF predicts labels from the scale-invariant score

$$s_{ik} = \|u_k\|_2^2 v_{ik}^2, \qquad \widehat{y}_i = \arg\max_k s_{ik}. \tag{18}$$

The same permutation-aligned ACC definition is used for every method in the external table. NMI is permutation invariant. The mechanism-specific ablation and sensitivity analyses additionally report the native, unpermuted PCE-NMF decision in Eq. (8).

Table 4: Locked settings for HSSNMF, CSNMF, and SPNMF on all six data sets.

Method	p	μ	α	β	λ	Iterations
HSSNMF	5	0.2	100	–	–	100
CSNMF	5	0.2	–	100	–	300
SPNMF	5	0.2	100	100	1000	100

5.2 Compared methods and parameter protocol

The external comparison includes NMF, CNMF, GNMFLD, EDNMF, LCNMF, GOCNMF, SNMFWLP, HSSNMF, CSNMF, SPNMF, and PCE-NMF. Standard NMF provides an unsupervised reference. CNMF uses the hard constraint $V = AZ$. GNMFLD combines labeled-coordinate fitting with graph regularization [16]. The EDNMF baseline uses its element-ratio variant, ERDNMF. LCNMF suppresses the wrong-class coordinates of labeled samples. GOCNMF fixes labeled representations to one-hot vectors and uses graph smoothness. SNMFWLP combines an orthogonal basis constraint with weighted label propagation. HSSNMF propagates pairwise constraints through a hypergraph, CSNMF combines pointwise and pairwise supervision with a correntropy loss, and SPNMF jointly propagates labels and pairwise constraints.

NMF, CNMF, GNMFLD, EDNMF, and LCNMF are run for 200 multiplicative-update iterations. GOCNMF, SNMFWLP, and PCE-NMF are run for 50 iterations; HSSNMF, CSNMF, and SPNMF use 100, 300, and 100 iterations, respectively, following their source-aligned implementations. PCE-NMF uses $\lambda = 100$ on all data sets; this value was fixed before the formal 20-seed comparison.

The neighborhood sizes and method-specific iteration budgets were frozen in the protocol before the 20-seed comparison. The formal table reports this pre-specified configuration, and Fig. 3 presents the sensitivity analysis separately. GNMFLD uses $p = 5$ throughout. PCE-NMF, GOCNMF, and SNMFWLP use the values listed in Table 2; shared values are used when the verified implementations admit the same graph. The EDNMF and LCNMF runs use preserved author-code snapshots, while the remaining methods use audited local adapters. Every transferred setting is identified in Table 3.

Table 3 reports the data-set-specific parameters. Entries marked T are fixed values transferred from a related source data set.

The three additional propagation baselines use one locked configuration on all six data sets. Table 4 records every parameter used by the corresponding adapters. Here, p is the neighborhood size and μ is the propagation coefficient. HSSNMF uses the common 100-iteration protocol cap; the remaining iteration budgets follow the source-aligned settings recorded before the formal runs.

For statistical comparison, we form paired differences between PCE-NMF and each baseline for every data-set–metric combination. A two-sided exact paired Wilcoxon signed-rank test [17] is applied to the 20 paired trials. Holm correction is performed within each pre-specified baseline family of 12 tests (six data sets times two metrics). We report the magnitude, direction, and adjusted p-value of each paired difference.

All computations use MATLAB R2019a on Windows with double-precision arrays. The reproducibility bundle records the 20 seeds, split-generation rule, method order, implementation provenance, transferred-parameter tags, raw trial records, summaries, paired tests, and diagnostics. The formal experiment contains $6 \times 20 \times 11 = 1320$ valid method–data-set–seed runs.

5.3 Overall clustering results

Table 5 reports the mean and standard deviation over the 20 paired trials. PCE-NMF ranks within the top three in all 12 data-set–metric combinations and is the best method on both metrics for COIL20 and COIL100. On COIL20, it reaches 0.9245 ACC and 0.9397 NMI, compared with 0.9179 and 0.9339 for GOCNMF. On COIL100, the corresponding results are 0.8666 and 0.9225, exceeding GOCNMF by 0.0154 ACC and 0.0098 NMI.

PCE-NMF exceeds GNMFLD on both metrics for YaleB, COIL20, ORL, and COIL100. The paired mean ACC improvements are 0.0838, 0.0388, 0.0296, and 0.0471, respectively. On Optdigits, PCE-NMF, GOCNMF, and GNMFLD all obtain approximately 0.984 ACC and 0.959 NMI. The PCE-NMF-GNMFLD differences are -2.77×10^{-4} ACC and -3.93×10^{-4} NMI and are statistically indistinguishable after Holm correction.

GOCNMF exceeds PCE-NMF on YaleB by 0.0334 ACC and 0.0279 NMI and on PIE by 0.0069 ACC and 0.0060 NMI. On ORL, PCE-NMF improves ACC by 0.0089, while GOCNMF improves NMI by 0.0059.

Against NMF, CNMF, EDNMF, LCNMF, HSSNMF, CSNMF, and SPNMF, PCE-NMF achieves higher mean ACC and NMI on all six data sets, with all corresponding Holm-adjusted tests significant. PCE-NMF also exceeds SNMFWLP on five data sets for both metrics. On Optdigits, the mean improvements are 0.0428 ACC and 0.0128 NMI, with adjusted $p = 0.158$ for both metrics.

5.4 Paired comparison with the strongest direct baselines

Table 6 focuses on GNMFLD and GOCNMF, the two strongest direct-assignment baselines in the overall experiment. Differences are PCE-NMF minus the stated baseline.

Table 7 gives the paired results for the three additional propagation baselines. PCE-NMF has a positive mean difference in all 36 comparisons. The seed-level win rate ranges from 0.85 to 1.00, and every comparison remains significant after Holm correction within its 12-test baseline family.

Figure 2 visualizes the same paired comparison with two-sided 95% Student confidence intervals for the mean difference across the 20 matched seeds. The zero line separates improvements from degradations. Gains over GNMFLD are largest on YaleB and COIL100. Differences among PCE-NMF, GNMFLD, and GOCNMF are close to zero on Optdigits, while ORL exhibits opposite ACC and NMI directions relative to GOCNMF.

On Optdigits, the differences between PCE-NMF and GOCNMF are below 5×10^{-4}. On ORL, PCE-NMF attains the highest ACC, whereas GOCNMF attains the highest

Table 3: Locked data-set-specific parameters. Entries marked T are transferred from a related source data set.

Data set	PCE λ	GOC α	GNMFLD α	GNMFLD β	ERDNMF α	Transfer notes
YaleB	100	1000	1000	50	10	GNMFLD, ERDNMF, SNMFWLP: T
COIL20	100	10	800	200	800	none
Optdigits	100	10	600	100	600	GNMFLD and SNMFWLP: T
ORL	100	100	400	10	10	GOC and ERDNMF: T
PIE	100	1000	1000	50	400	GNMFLD and SNMFWLP: T
COIL100	100	10	600	50	1000	none

Table 5: Clustering performance (mean $\pm$ standard deviation) over 20 paired trials. ACC is permutation-aligned according to the definition above, and both ACC and NMI are evaluated only on unlabeled samples. Bold and underlined entries denote the best and second-best mean values, respectively.

Data set	Metric	NMF	CNMF	GNMFLD	EDNMF	LCNMF	GOCNMF	SNMFWLP	HSSNMF	CSNMF	SPNMF	PCE-NMF
YaleB	ACC	0.1626 ± 0.0111	0.1636 ± 0.0111	0.6150 ± 0.0127	0.2228 ± 0.0125	0.2576 ± 0.0123	$\mathbf{0.7321 \pm 0.0148}$	0.5536 ± 0.0159	0.2530 ± 0.0132	0.1515 ± 0.0136	0.4268 ± 0.0283	$\underline{0.6988 \pm 0.0158}$
	NMI	0.3007 ± 0.0098	0.3032 ± 0.0109	0.6243 ± 0.0097	0.3756 ± 0.0112	0.3828 ± 0.0105	$\mathbf{0.7336 \pm 0.0099}$	0.6331 ± 0.0140	0.3978 ± 0.0133	0.2256 ± 0.0168	0.5325 ± 0.0337	$\underline{0.7057 \pm 0.0104}$
COIL20	ACC	0.6268 ± 0.0265	0.6483 ± 0.0322	0.8857 ± 0.0086	0.6942 ± 0.0152	0.6898 ± 0.0124	$\underline{0.9179 \pm 0.0090}$	0.6687 ± 0.0309	0.8112 ± 0.0261	0.7878 ± 0.0378	0.7088 ± 0.0389	$\mathbf{0.9245 \pm 0.0092}$
	NMI	0.7360 ± 0.0091	0.7501 ± 0.0181	0.9128 ± 0.0067	0.7594 ± 0.0137	0.7556 ± 0.0142	$\underline{0.9339 \pm 0.0075}$	0.7791 ± 0.0291	0.9267 ± 0.0070	0.9161 ± 0.0085	0.8503 ± 0.0247	$\mathbf{0.9397 \pm 0.0060}$
Optdigits	ACC	0.7062 ± 0.0413	0.7600 ± 0.0288	$\mathbf{0.9838 \pm 0.0016}$	0.8229 ± 0.0090	0.7696 ± 0.0672	$\underline{0.9837 \pm 0.0013}$	0.9407 ± 0.0548	0.9605 ± 0.0437	0.9423 ± 0.0479	0.9108 ± 0.0493	0.9835 ± 0.0013
	NMI	0.6524 ± 0.0163	0.6768 ± 0.0196	$\underline{0.9588 \pm 0.0032}$	0.7109 ± 0.0101	0.6788 ± 0.0443	$\mathbf{0.9588 \pm 0.0026}$	0.9456 ± 0.0176	0.9482 ± 0.0141	0.9432 ± 0.0131	0.9348 ± 0.0168	0.9584 ± 0.0027
ORL	ACC	0.5704 ± 0.0203	0.5725 ± 0.0316	$\underline{0.6257 \pm 0.0279}$	0.5500 ± 0.0201	0.5874 ± 0.0252	$\underline{0.6464 \pm 0.0243}$	0.5804 ± 0.0181	0.5112 ± 0.0164	0.4583 ± 0.0132	0.5413 ± 0.0342	$\mathbf{0.6553 \pm 0.0255}$
	NMI	0.7695 ± 0.0120	0.7709 ± 0.0142	0.7670 ± 0.0145	0.7030 ± 0.0161	0.7321 ± 0.0176	$\mathbf{0.7942 \pm 0.0148}$	0.7674 ± 0.0124	0.7404 ± 0.0075	0.6936 ± 0.0073	0.7198 ± 0.0254	$\underline{0.7883 \pm 0.0142}$
PIE	ACC	0.5343 ± 0.0227	0.5574 ± 0.0238	$\mathbf{0.9426 \pm 0.0060}$	0.7736 ± 0.0169	0.7608 ± 0.0094	$\underline{0.9396 \pm 0.0112}$	0.5537 ± 0.0292	0.7727 ± 0.0249	0.7403 ± 0.0273	0.3271 ± 0.0261	0.9327 ± 0.0111
	NMI	0.7961 ± 0.0081	0.8038 ± 0.0100	$\underline{0.9512 \pm 0.0036}$	0.8500 ± 0.0101	0.8410 ± 0.0037	$\mathbf{0.9554 \pm 0.0067}$	0.7550 ± 0.0252	0.9193 ± 0.0053	0.9115 ± 0.0061	0.5417 ± 0.0239	0.9494 ± 0.0059
COIL100	ACC	0.4633 ± 0.0129	0.4830 ± 0.0095	$\underline{0.8195 \pm 0.0061}$	0.5519 ± 0.0115	0.5458 ± 0.0086	$\underline{0.8512 \pm 0.0080}$	0.4122 ± 0.0150	0.6777 ± 0.0153	0.6737 ± 0.0217	0.1953 ± 0.0123	$\mathbf{0.8666 \pm 0.0053}$
	NMI	0.7413 ± 0.0053	0.7470 ± 0.0036	0.8952 ± 0.0024	0.7238 ± 0.0063	0.7210 ± 0.0049	$\underline{0.9127 \pm 0.0040}$	0.6286 ± 0.0177	0.8915 ± 0.0036	0.8928 ± 0.0063	0.3808 ± 0.0148	$\mathbf{0.9225 \pm 0.0021}$

Table 6: Paired comparisons with GNMFLD and GOCNMF. Values are mean paired differences; parentheses contain Holm-adjusted p-values.

	PCE $-$ GNMFLD		PCE $-$ GOCNMF	
Data set	ACC	NMI	ACC	NMI
YaleB	$+0.08381$ (2.29×10^{-5})	$+0.08130$ (2.29×10^{-5})	-0.03338 (2.29×10^{-5})	-0.02792 (2.29×10^{-5})
COIL20	$+0.03885$ (2.29×10^{-5})	$+0.02698$ (2.29×10^{-5})	$+0.00662$ (0.0126)	$+0.00585$ (0.0126)
Optdigits	-0.00028 (0.302)	-0.00039 (0.302)	-0.00017 (0.0247)	-0.00041 (0.0247)
ORL	$+0.02958$ (3.72×10^{-4})	$+0.02132$ (2.17×10^{-4})	$+0.00889$ (0.00373)	-0.00589 (0.00373)
PIE	-0.00998 (5.34×10^{-4})	-0.00178 (0.302)	-0.00693 (3.05×10^{-5})	-0.00604 (3.05×10^{-5})
COIL100	$+0.04710$ (2.29×10^{-5})	$+0.02725$ (2.29×10^{-5})	$+0.01541$ (2.29×10^{-5})	$+0.00981$ (2.29×10^{-5})

Table 7: Paired comparisons with HSSNMF, CSNMF, and SPNMF. Each entry is the mean difference PCE-NMF minus baseline; parentheses contain the Holm-adjusted p-value.

Data set	Metric	PCE $-$ HSSNMF	PCE $-$ CSNMF	PCE $-$ SPNMF
YaleB	ACC	$+0.44577$ (2.29×10^{-5})	$+0.54728$ (2.29×10^{-5})	$+0.27197$ (2.29×10^{-5})
	NMI	$+0.30790$ (2.29×10^{-5})	$+0.48003$ (2.29×10^{-5})	$+0.17315$ (2.29×10^{-5})
COIL20	ACC	$+0.11327$ (2.29×10^{-5})	$+0.13673$ (2.29×10^{-5})	$+0.21569$ (2.29×10^{-5})
	NMI	$+0.01301$ (5.34×10^{-5})	$+0.02363$ (2.29×10^{-5})	$+0.08940$ (2.29×10^{-5})
Optdigits	ACC	$+0.02303$ (5.34×10^{-5})	$+0.04127$ (2.29×10^{-5})	$+0.07278$ (2.67×10^{-4})
	NMI	$+0.01024$ (2.29×10^{-5})	$+0.01521$ (2.29×10^{-5})	$+0.02357$ (2.67×10^{-4})
ORL	ACC	$+0.14403$ (2.29×10^{-5})	$+0.19694$ (2.29×10^{-5})	$+0.11403$ (2.29×10^{-5})
	NMI	$+0.04789$ (2.29×10^{-5})	$+0.09471$ (2.29×10^{-5})	$+0.06850$ (2.29×10^{-5})
PIE	ACC	$+0.15997$ (2.29×10^{-5})	$+0.19236$ (2.29×10^{-5})	$+0.60551$ (2.29×10^{-5})
	NMI	$+0.03007$ (2.29×10^{-5})	$+0.03792$ (2.29×10^{-5})	$+0.40770$ (2.29×10^{-5})
COIL100	ACC	$+0.18888$ (2.29×10^{-5})	$+0.19288$ (2.29×10^{-5})	$+0.67129$ (2.29×10^{-5})
	NMI	$+0.03103$ (2.29×10^{-5})	$+0.02966$ (2.29×10^{-5})	$+0.54169$ (2.29×10^{-5})

NMI.

5.5 Parameter sensitivity

We perform a sensitivity analysis on YaleB, COIL20, and COIL100 using the first 10 locked seeds. We vary $\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}$ at the locked neighborhood size, and vary $p \in \{2, 3, 4, 5, 6\}$ at $\lambda = 100$. All other splits, initializations, updates, and unlabeled-only evaluations are unchanged. Figure 3 reports native PCE-NMF ACC and NMI; open circles mark the parameter values used in the formal comparison.

Performance increases sharply between $\lambda = 0.01$ and

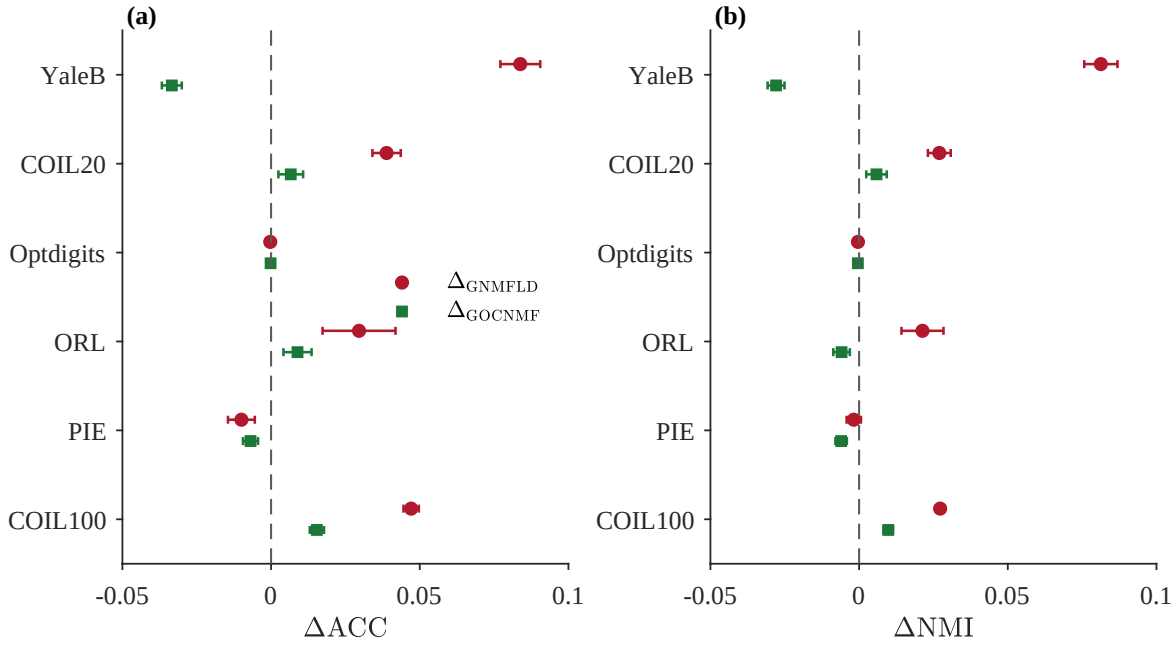


Figure 2: Paired performance differences of PCE-NMF relative to GNMFLD and GOCNMF. For baseline B and metric M, let $\Delta_B = M(\text{PCE-NMF}) - M(B)$, where $B \in \{\text{GNMFLD}, \text{GOCNMF}\}$ and $M \in \{\text{ACC}, \text{NMI}\}$. Panels (a) and (b) report ACC and NMI, respectively. Markers and error bars show the mean and the interval $\overline{\Delta}_B \pm t_{0.975,19} s_\Delta / \sqrt{20}$ over the matched seeds. The dashed line marks $\Delta_B = 0$, and positive values favor PCE-NMF.

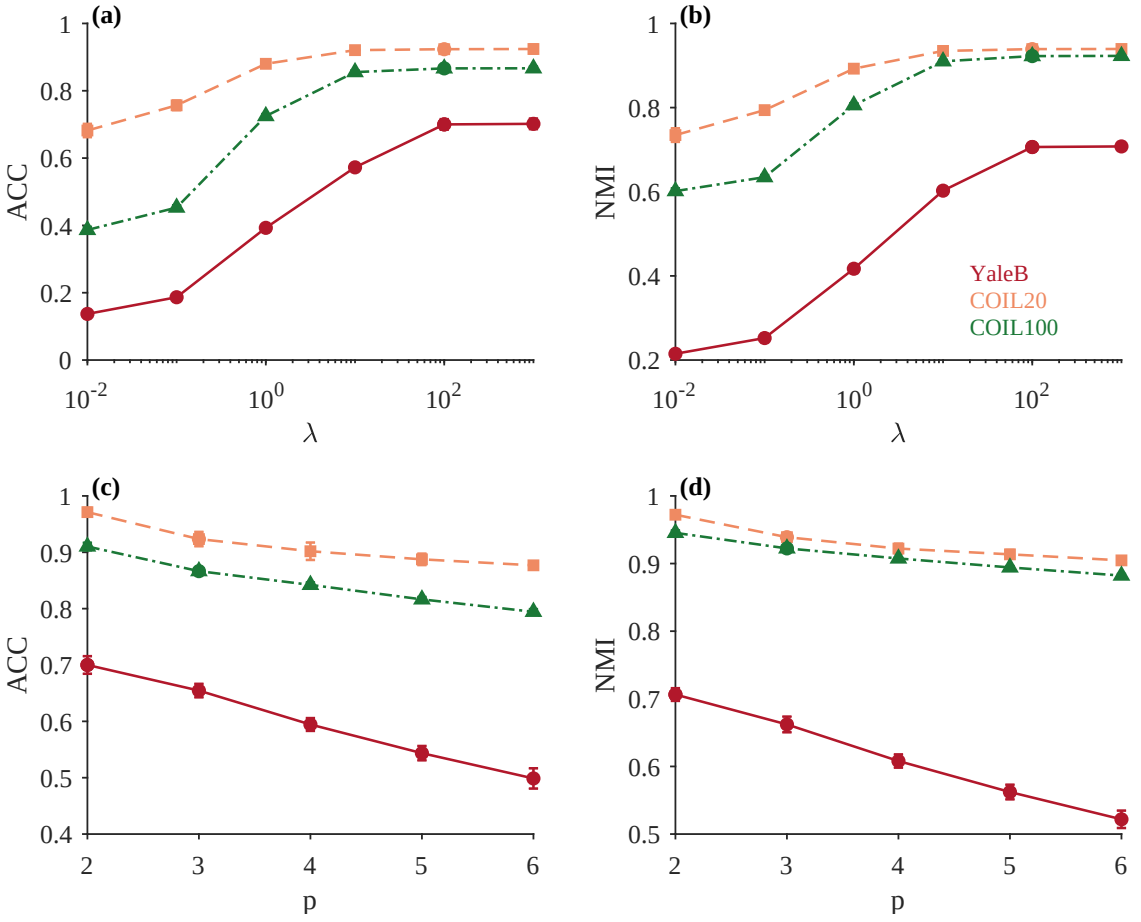


Figure 3: Parameter sensitivity of PCE-NMF over 10 paired seeds. (a,b) ACC and NMI versus λ at the locked p; (c,d) ACC and NMI versus p at $\lambda = 100$. Error bars show one standard deviation and open circles mark the settings used in the 20-seed comparison.

10, and changes only slightly from $\lambda = 100$ to 1000. At $\lambda = 100$, the mean direct ACC values are 0.7000, 0.9235, and 0.8665 on YaleB, COIL20, and COIL100; at $\lambda = 1000$ they are 0.7017, 0.9241, and 0.8667. The setting $\lambda = 100$ therefore lies on a broad performance plateau.

The neighborhood size is more influential. YaleB attains

its highest tested mean at the formal $p = 2$. On COIL20 and COIL100, $p = 2$ results exceed the formal $p = 3$ results: direct ACC changes from 0.9235 to 0.9712 on COIL20 and from 0.8665 to 0.9103 on COIL100. These results show that graph-neighborhood size materially affects performance.

5.6 Influence of supervisory information

Because PCE-NMF is designed to propagate limited class evidence, we further investigate its behavior under different amounts of supervision. YaleB, COIL20, and COIL100 provide one face data set and two object data sets with 20 and 100 classes, respectively, and therefore expose both data-type and class-count changes within a tractable repeated experiment. The labeled proportion per class is varied over $\{5\%, 10\%, 15\%, 20\%\}$ using nested labeled sets, while a common evaluation pool is retained for all proportions within each paired trial. Figure 4 reports the mean ACC and NMI over 20 paired seeds.

The benefit of propagated class exclusion is most pronounced in the low-supervision regime. On COIL20 with only 5% labeled samples, PCE-NMF achieves an ACC of 0.8966, compared with 0.8274 for GOCNMF, while the corresponding NMI values are 0.9296 and 0.8620. The advantage is even more evident on COIL100, where PCE-NMF improves ACC from 0.7063 to 0.8121 and NMI from 0.8312 to 0.9023.

As more labeled samples become available, the gap between PCE-NMF and GOCNMF gradually decreases. The two methods become nearly tied at 15%-20% supervision on COIL20 and at 20% on COIL100. This behavior is consistent with the role of harmonic class evidence: propagation provides the greatest additional information when direct supervision is scarce, whereas its marginal contribution naturally decreases as more labeled samples become available.

On YaleB, GOCNMF remains stronger over all four supervision levels, and PCE-NMF consistently outperforms GNMFLD and SNMFWLP. Together, these results locate the strongest PCE-NMF gains in low-supervision object-clustering problems with many classes.

5.7 Ablation of propagated class exclusion

To isolate the contribution of propagated class exclusion, we compare PCE-NMF with a hard-class-exclusion variant, denoted HCE-NMF. HCE-NMF preserves the same factorization structure and the labeled wrong-class zero constraints but sets the propagated-exclusion regularization weight to zero. This comparison measures the incremental effect of the complete propagated-exclusion term over the hard-support model.

Both variants have class-semantic latent coordinates and are evaluated using the native class-energy decision

$$\widehat{y}_i = \arg\max_k \|u_k\|_2^2 v_{ik}^2, \tag{19}$$

without permutation alignment. Both variants use the same seeds, labeled splits, initial factors, cosine graphs, updates, and NMI definition. Table 8 reports the ablation results over the 20 paired trials.

PCE-NMF improves both metrics on every data set. The mean ACC gains over HCE-NMF range from 0.2593 on COIL20 to 0.5883 on PIE, while the NMI gains range from 0.2195 to 0.4944. PCE-NMF wins all 120 seed-level comparisons for each metric. All 12 data-set–metric tests remain significant after Holm correction (adjusted $p = 2.29\times10^{-5}$). These results show that hard compatibility constraints alone are insufficient in this model: propagation supplies useful class-incompatibility information for unlabeled samples.

Table 8: Ablation of propagated class exclusion under the native direct class-energy decision. Values are mean ± standard deviation over 20 paired trials.

	HCE-NMF		PCE-NMF	
Data set	ACC	NMI	ACC	NMI
YaleB	0.1319 ± 0.0117	0.2113 ± 0.0117	0.6988 ± 0.0158	0.7057 ± 0.0104
COIL20	0.6649 ± 0.0196	0.7203 ± 0.0162	0.9242 ± 0.0100	0.9397 ± 0.0060
Optdigits	0.6386 ± 0.1018	0.5912 ± 0.0552	0.9835 ± 0.0013	0.9584 ± 0.0027
ORL	0.2725 ± 0.0228	0.5075 ± 0.0186	0.6428 ± 0.0286	0.7883 ± 0.0142
PIE	0.3444 ± 0.0173	0.5147 ± 0.0160	0.9327 ± 0.0111	0.9494 ± 0.0059
COIL100	0.3754 ± 0.0086	0.5968 ± 0.0065	0.8652 ± 0.0053	0.9225 ± 0.0021

Figure 5 traces this mechanism in a formal COIL20 trial. The harmonic evidence concentrates near the correct class coordinate, its complement assigns small exclusion weight to that coordinate and large weights to incompatible coordinates, and the learned class-energy share becomes strongly diagonal. Thus the observed gain is accompanied by the matrix pattern predicted by the PCE penalty.

5.8 Visualization of learned factors

To further examine the representations learned by the competing methods, we visualize their basis vectors on COIL20. Since NMF admits an inherent diagonal scaling ambiguity, directly comparing the raw basis matrices may be misleading. Therefore, each learned basis column is normalized to unit Euclidean norm before visualization. For PCE-NMF, this normalization corresponds exactly to the canonical representation $B = UD_u^{-1}$ introduced in Section 4.2, where $\|b_k\|_2 = 1$.

Figure 6 compares the canonical basis vectors learned in one locked COIL20 trial under the same labeled split and initialization. Each basis vector is reshaped into a 32×32 image, and a common grayscale range is used for all four methods. In this trial, the PCE-NMF basis images appear smooth and spatially coherent, whereas the SNMFWLP row is more fragmented. The GNMFLD and GOCNMF rows exhibit intermediate spatial variation.

To describe the spatial regularity observed in Fig. 6, we additionally compute the spatial total variation of each normalized basis image, defined as the sum of the absolute horizontal and vertical differences between adjacent pixels. Averaged over the 20 basis vectors, the resulting values are 7.9290, 12.2434, 20.6700, and 6.9216 for GNMFLD, GOCNMF, SNMFWLP, and PCE-NMF, respectively. PCE-NMF therefore attains the lowest mean total variation in this trial, consistent with the smoother and more spatially coherent basis images in Fig. 6.

The learned factors combine the canonical basis with the class-specific contribution energies of PCE-NMF. The canonical basis captures the spatial structure of the data, whereas the propagated class-exclusion weights regulate the class-specific contribution energies used for direct assignment. Together, the normalized basis images and contribu-

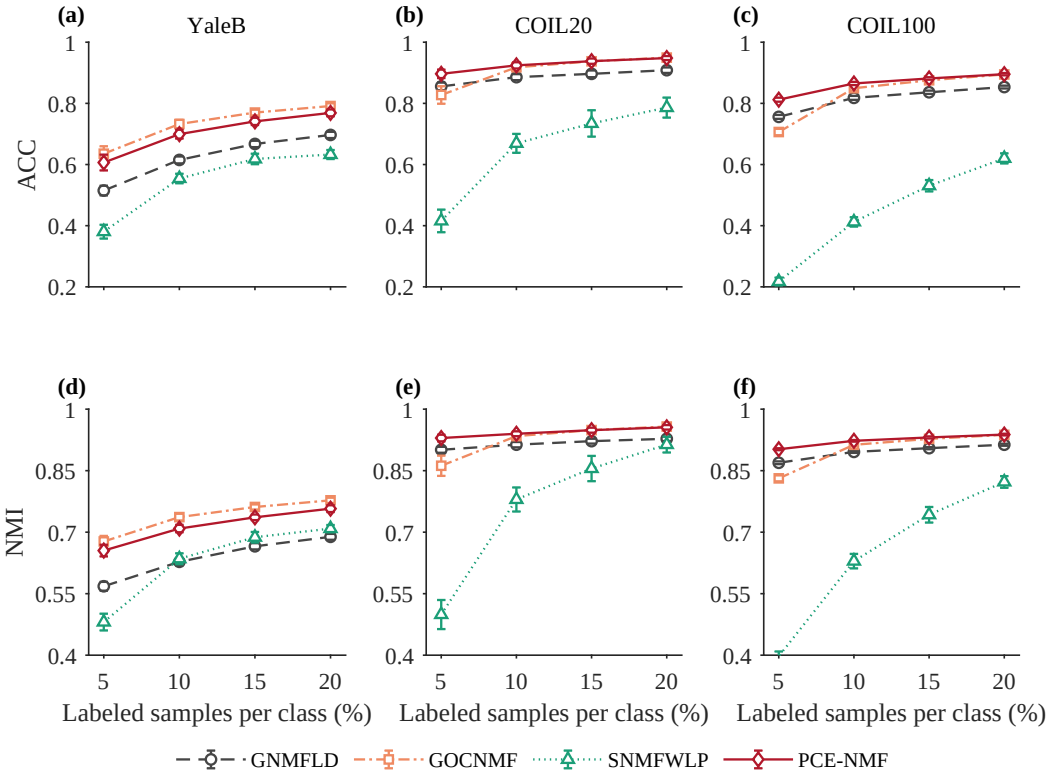


Figure 4: Influence of the proportion of labeled samples on clustering performance. The upper row reports ACC and the lower row reports NMI on YaleB, COIL20, and COIL100. For each paired seed, the labeled sets are nested across 5%, 10%, 15%, and 20% supervision, while all settings share a common evaluation pool. Markers denote the mean results over 20 paired trials, and error bars indicate one standard deviation.

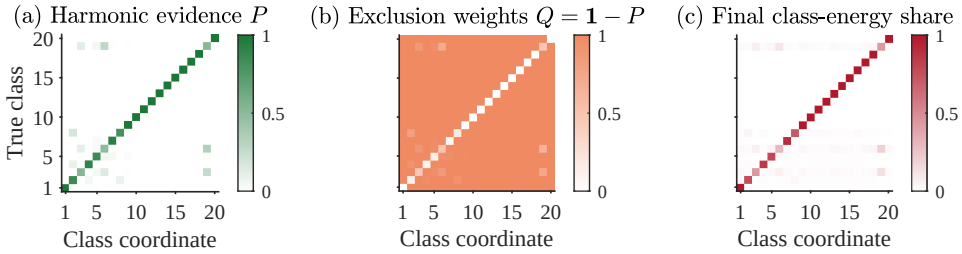


Figure 5: Propagation-to-factorization mechanism on COIL20 for seed 20260617, $p = 3$, $\lambda = 100$, and 50 iterations. Each row averages all unlabeled samples of one true class. (a) Harmonic class evidence P; (b) propagated exclusion weights $Q = 1 - P$; and (c) the final normalized class-energy share $s_{ik} / \sum_j s_{ij}$.

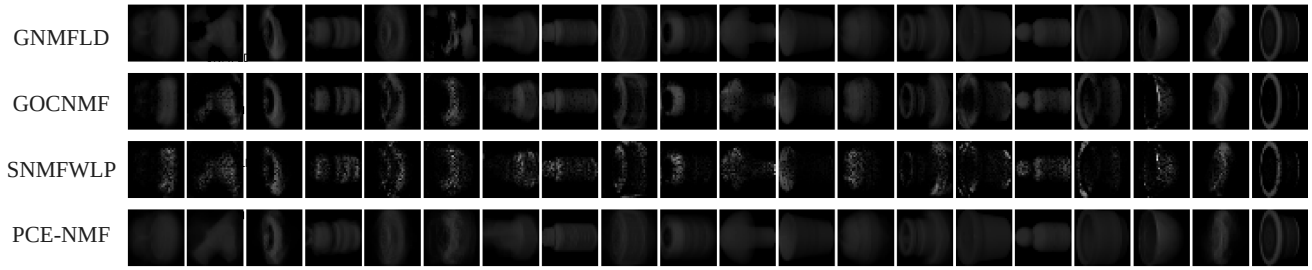


Figure 6: Canonical basis vectors from one locked COIL20 trial under a common labeled split and initialization. Rows correspond, from top to bottom, to GNMFLD, GOCNMF, SNMFWLP, and PCE-NMF. Each basis vector is normalized to unit Euclidean norm and reshaped into a 32×32 image; all rows use the same grayscale range.

tion energies show the spatial representation and the class-assignment mechanism learned by PCE-NMF.

5.9 Empirical convergence behavior

We record the PCE-NMF objective for 50 iterations on YaleB, COIL20, and COIL100 using the first 10 locked seeds and the formal parameters. Figure 7 shows the mean normalized

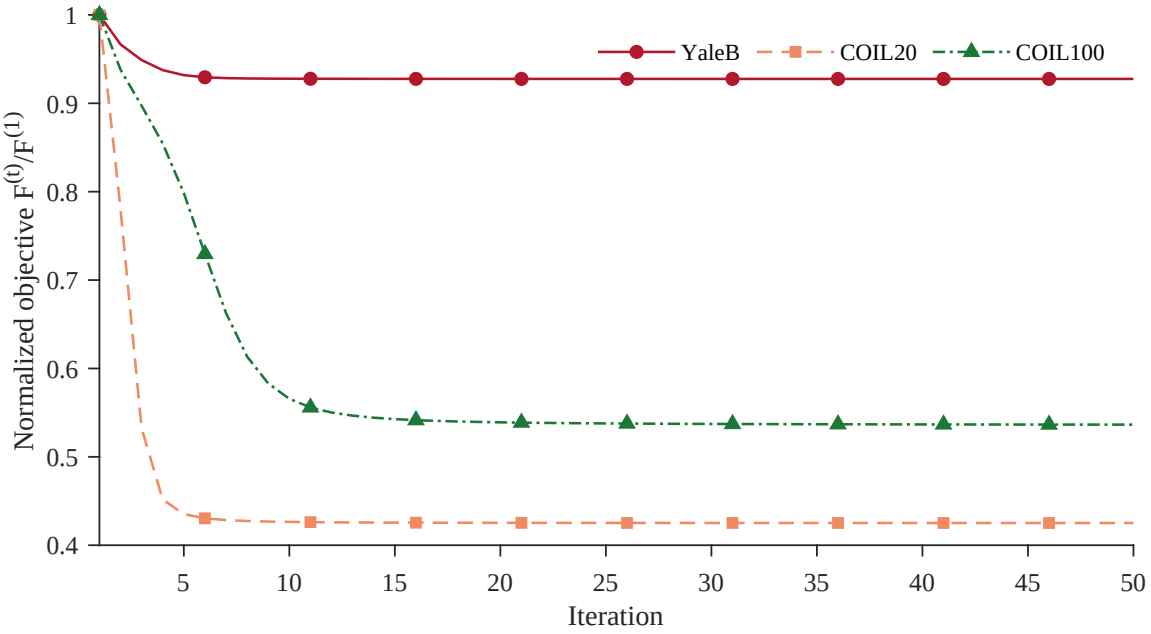


Figure 7: Mean objective trajectories of PCE-NMF over 10 paired seeds at the formal settings. Curves show $F^{(t)}/F^{(1)}$; sparse markers distinguish the three data sets.

objective $F^{(t)}/F^{(1)}$.

All 30 recorded trajectories are finite and nonincreasing under the numerical tolerance used by the implementation. At iteration 50, the mean ratios $F^{(50)}/F^{(1)}$ are 0.9276, 0.4252, and 0.5364 on YaleB, COIL20, and COIL100, respectively. The corresponding mean relative changes in the final step are 1.60×10^{-7}, 3.39×10^{-6}, and 3.46×10^{-5}. These measurements establish objective-value descent for the recorded runs.

5.10 Propagation coverage and constraint diagnostics

The harmonic field is defined component-wise, and propagated class-exclusion weights are introduced only on connected components that contain at least one labeled anchor. This condition is important because an unanchored component has no label information from which class evidence can be inferred. In such a component, PCE-NMF sets the propagated-exclusion weights to zero rather than introducing artificial supervision.

Across the 20 paired trials, the mean anchor coverage is 1.000 on YaleB, COIL20, and Optdigits; 0.937 on ORL; 0.996 on PIE; and 0.994 on COIL100. The corresponding minimum coverage values are 1.000, 1.000, 1.000, 0.914, 0.977, and 0.992, respectively. Thus, all unlabeled samples are anchored on YaleB, COIL20, and Optdigits, while only a small fraction of samples belongs to unanchored components on ORL, PIE, and COIL100. In particular, the minimum observed coverage remains above 0.91 on every data set. These results show that the harmonic evidence reaches the great majority of unlabeled samples under the graph settings used in the formal experiments, while the component-wise construction prevents unsupported class evidence from being assigned to disconnected regions.

We further examine numerical feasibility of the constrained factorization. All 1320 method runs return finite factors and finite evaluation metrics. For PCE-NMF, the labeled wrong-class entries are required to remain exactly zero throughout optimization. The maximum measured wrong-class support violation is zero over all PCE-NMF runs; the same holds for LCNMF, which also imposes hard wrong-class support constraints. Hence, the implemented multiplicative updates preserve the prescribed support structure to numerical precision. Together, the coverage and feasibility diagnostics confirm that the propagated evidence is applied only where labeled information is available and that the hard exclusion constraints are maintained consistently during optimization.

5.11 Computational cost

Table 9 reports average recorded trial time. Common data loading, normalization, and precomputed dataset-level graphs are excluded; method-specific operations executed inside an adapter, including clustering assignment and propagation, are included. The PCE-NMF time includes its split-dependent harmonic propagation and factorization. Under this protocol, PCE-NMF requires between 0.663 and 5.662 seconds per trial. It is the fastest method on five data sets and the third fastest on ORL.

5.12 Discussion

The experiments separate two sources of supervision. Hard support constraints assign class semantics to labeled coordinates, while propagated exclusion extends class-incompatibility information to unlabeled samples. Removing the second source produces large losses on every data set and every paired seed. The diagonal patterns in Fig. 5 connect this performance change to the intended sequence from harmonic evidence to exclusion weights and class-energy concentration.

The external comparison places PCE-NMF among the top

Table 9: Average recorded trial time in seconds. Common data loading, normalization, and precomputed dataset-level graphs are excluded, whereas method-specific adapter operations are included. PCE-NMF includes split-dependent harmonic propagation and factorization.

Data set	NMF	CNMF	GNMFLD	EDNMF	LCNMF	GOCNMF	SNMFWLP	HSSNMF	CSNMF	SPNMF	PCE-NMF
YaleB	8.317	7.666	5.508	4.408	4.139	1.310	5.263	4.699	54.071	27.044	**1.229**
COIL20	2.977	3.082	2.844	2.827	2.404	0.716	1.318	1.149	20.441	8.049	**0.663**
Optdigits	4.254	4.086	6.696	2.122	1.040	1.532	3.404	103.815	276.651	103.569	**0.994**
ORL	4.660	4.812	4.428	4.360	4.183	1.110	1.662	**0.780**	14.740	4.864	1.220
PIE	15.078	14.894	10.107	7.745	6.827	2.283	10.450	45.219	91.109	36.091	**2.005**
COIL100	50.145	46.728	31.883	21.790	16.323	7.161	62.416	224.287	1089.139	247.635	**5.662**

three methods in every data-set–metric combination. Its mean improvements over HSSNMF, CSNMF, and SPNMF are positive in all 36 data-set–metric comparisons and remain significant after within-baseline Holm correction. The comparison with GOCNMF reveals a data-dependent choice between positive specification and exclusion. GOCNMF leads on YaleB and PIE, PCE-NMF leads on COIL20 and COIL100, ORL splits between ACC and NMI, and Optdigits produces practically equal results. The label-rate experiment sharpens this distinction: PCE-NMF gains most strongly on the multi-class object data when labeled evidence is scarce, and the gap contracts as direct supervision increases.

Graph construction controls how far class evidence travels. The λ curves place $\lambda = 100$ on a broad plateau, whereas the neighborhood curves show large changes from $p = 3$ to $p = 2$ on COIL20 and COIL100. This result makes p a central model-selection variable. Label-only graph calibration is therefore the primary deployment-stage control, and the formal comparison uses the pre-specified neighborhood values reported in Table 2.

The objective trajectories, anchor-coverage measurements, and zero-support diagnostics connect the mathematical model to the executed algorithm. They show monotonic finite-precision descent in all recorded trajectories, broad reach of the harmonic evidence, and exact preservation of labeled wrong-class zeros.

The data-set pattern distinguishes the regimes favored by the two strongest mechanisms. GOCNMF leads on YaleB and PIE, the strongest direct-assignment methods converge to nearly identical performance on Optdigits, and PCE-NMF shows its largest gains on COIL20 and COIL100 under low supervision. Across the three label-rate data sets, this advantage decreases as direct supervision increases. Neighborhood size has a larger effect than λ on COIL20 and COIL100, establishing graph calibration as the principal model-selection step. The blocked neighbor search and sparse harmonic solve provide the computational route for extending PCE-NMF beyond the largest collection tested here ($n = 7200$).

6 Conclusion

PCE-NMF converts harmonic class evidence into a class-incompatibility field and applies it to diagonal-scale-invariant rank-one contribution energies. The same energies define a direct class decision, linking supervision, factorization, and assignment through one mathematical quantity. The canonical representation expresses this mechanism as sample- and class-specific quadratic shrinkage, and the ideal alternating updates generate a non-increasing objective sequence on the stated positive domain. Across six benchmark data sets and eleven methods, PCE-NMF ranks within the top three for every data-set–metric combination and achieves the highest mean ACC and NMI on COIL20 and COIL100. It produces higher mean ACC and NMI than HSSNMF, CSNMF, and SPNMF on all six data sets, with all paired comparisons significant after within-baseline Holm correction. The HCE-NMF ablation shows substantial gains from the complete propagated-exclusion term beyond hard class support. Label-rate experiments locate the strongest observed benefit in low-supervision multi-class object clustering, while neighborhood sensitivity identifies graph calibration as the main operational control. Propagated class exclusion provides a direct and interpretable route from partial labels to class-semantic NMF clustering.

CRediT authorship contribution statement

Deshu Sun: Writing–original draft, Methodology, Software, Conceptualization, Formal analysis.
Feng Wang: Writing–review and editing, Funding acquisition, Visualization, Conceptualization. All authors reviewed the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The supplementary reproducibility archive contains the MATLAB R2019a implementation, locked configurations, seed and split rules, method-source notes, raw trial records, summary tables, paired tests, diagnostics, and the source data and scripts used for the reported figures. The six benchmark data sets are third-party resources and are not redistributed as part of the public archive. Their locked filenames, dimensions, label ranges, preprocessing, and access notes are recorded in the accompanying data-source manifest; use of each data set remains subject to the terms of its original provider.

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